

LESSON 1.4

OPERATIONS WITH INTEGERS



LESSON INTRODUCTION

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.

LEARNING TARGETS

- I can add, subtract, multiply, and divide integers.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

ESSENTIAL QUESTIONS

- How can I apply my knowledge of integers to determine rules for adding, subtracting, multiplying, and dividing with positives and negatives?

LESSON NARRATIVE

This lesson extends students' understanding of integers from 6th grade to be able to add, subtract, multiply, and divide integers. Students start by using integer chips and number line models to add and subtract integers and rediscover the rules for adding and subtracting with integers. Next, students use number lines and integer chips to help discover the rules for multiplying and dividing integers.

COMPONENT SUMMARY

1.4.1 WARM-UP (~5 MIN.)

Critical thinking Warm-Up

1.4.3 GUIDED PRACTICE (10 MIN.)

Adding and subtracting integers

1.4.5 GUIDED PRACTICE (15 MIN.)

Multiplying and dividing integers rules

1.4.7 WRAP-UP (~5 MIN.)

Self-reflection on the lesson

1.4.2 ACTIVITY (10 MIN.)

Integers Card Sort

1.4.4 BRING IT TOGETHER (5 MIN.)

Adding and subtracting integers rules

1.4.6 TRY IT (10 MIN.)

Practice with operations with integers

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Inverse
- Zero pair

MATERIALS

- Card Sort
- (Optional) Integer chips
- (Optional) Blank notecards

ADDITIONAL PREPARATION

- (Optional) Visual of rules for students to reference

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may subtract a negative integer as the addition of a negative integer rather than taking the “opposite of a negative” integer. Have students compare the 5th and 8th numeric expression in the Card Sort.

THINGS TO CONSIDER

- The lesson reviews operations with integers. Adapt the lesson to the gaps and needs of your students based on their understanding of the operations with integers, including adjusting times as necessary based on student understanding.
- Utilize Grade 6 Unit 7 resources for additional scaffolding, videos, and models.
- If time is an issue, consider assigning 1.4.6 Try It for homework.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 1.4.3 Guided Practice, consider implementing a Notice and Wonder routine. Prompt students to think about the following: What do you notice about the sum when you subtract a negative number? Why do you think that is? What does that make you wonder?

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Algebraic descriptions, number lines, and integer chips – students solidify their understanding of equivalence using multiple representations of a mathematical expression. Encourage students to substitute words like “same” with “equivalent” or “picture” with “model.”

DIFFERENTIATE INSTRUCTION

Discussion provides additional entry points for auditory learners. Consider utilizing sentence stems in 1.4.5 Guided Practice, and provide ample time for students to grapple with academic language:

The expression $(-2) \cdot 4$ and $2 \cdot -4$ are different because _____ this makes me wonder _____

Additional differentiation opportunities are denoted throughout the Lesson Guide.

1.4.1 WARM-UP: CRITICAL THINKING

ENRICHMENT

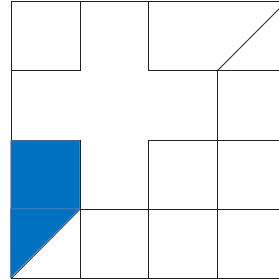
Consider prompting questions to analyze student thinking for identifying different methods and approaches to highlight for the class.

- What did you look for first?
- How did you determine “how much” of the entire design was shaded?
- What was easy? What was hard?



1.4.1 Warm-Up: Critical Thinking

1. A design is shown, where one part of the design is shaded.



- a. What fraction of the diagram is shaded?

$$\frac{3}{32}$$

- b. Explain how you determined your answer.

Answers will vary. The small square is $\frac{1}{16}$ of the larger square. This makes the shaded triangle $\frac{1}{32}$ of the larger square since it is half of the shaded square. $\frac{1}{16} + \frac{1}{32} = \frac{3}{32}$

1.4.2 ACTIVITY: CARD SORT

QUESTIONING

To challenge students, consider taking away some cards and have students create the missing cards on blank notecards.

DIFFERENTIATE INSTRUCTION

For additional scaffolding regarding number lines and integer chips, utilize models, visuals, and Study Expert videos from Grade 6 Unit 7 resources.

TEACHER MOVES

Students should be collaborating with their group throughout the component. Monitor and listen for common student misconceptions in order to identify which students surface those misconceptions and which students help clarify.



1.4.2 Activity: Card Sort

1. Work with your partner or group to match each expression with the integer chip model and number line model that represents the expression. Record your answers in the table. Then, use the models to determine the value of each expression.

Mathematical Expression	Letter of Integer Chip Model	Letter of Number Line Model	Value
$4 + (-7)$	<i>P</i>	<i>M</i>	-3
$-3 + (-2)$	<i>X</i>	<i>C</i>	-5
$7 - 10$	<i>Y</i>	<i>N</i>	-3
$-5 - 3$	<i>D</i>	<i>B</i>	-8
$-8 - (-2)$	<i>K</i>	<i>V</i>	-6
$4 - (-6)$	<i>T</i>	<i>G</i>	10
$4 + (-5)$	<i>S</i>	<i>W</i>	-1
$-2 + (-8)$	<i>H</i>	<i>R</i>	-10

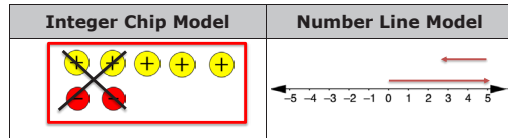
2. Explain the method you used to match the cards together.

Answers will vary. I used the chip model because the chips show opposites. It is much easier to make pairs of zero with the chip model.

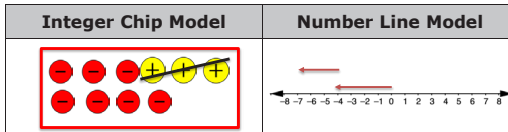


1.4.3 Guided Practice: Adding and Subtracting Integers

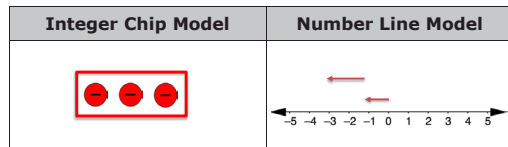
- Model adding or subtracting the two integers by using the integer chip model and the number line model.
 - $5 + (-2) = \underline{\quad 3 \quad}$



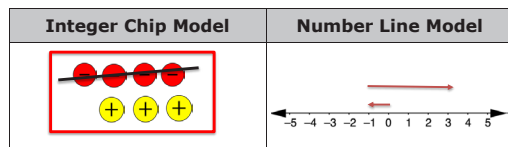
- $-4 - 3 = \underline{\quad -7 \quad}$



- $(-1) + (-2) = \underline{\quad -3 \quad}$



- $-1 - (-4) = \underline{\quad 3 \quad}$



- Complete the statements.

When adding integers of different signs using the integer chip model, the smaller number of chips will combine with the same number of chips of the opposite sign to form
 ◦ the sum.
 ◦ zero pairs.
 The remaining chips show
 ◦ the sum.
 ◦ zero
 The sum has the sign of the
 ◦ larger
 ◦ smaller
 number of chips.

- Complete the table.

Expression	Number of Zero Pairs Created	Sign of Sum	Sum
$(-9) + 12$	9	+	3
$13 + -7$	7	+	6
$(-10) + 5$	5	-	-5
$7 + (-7)$	7	Neither	0

1.4.3 GUIDED PRACTICE: ADDING AND SUBTRACTING INTEGERS

DIFFERENTIATE INSTRUCTION

Consider providing students with actual integer chips to model with. Students can practice manipulating the integer chips to add and subtract. Every time there is one negative and one positive, that makes a zero pair.

ENRICHMENT

How does your integer chip model change when you are adding a negative? How does your integer chip model change when you are subtracting a positive?

ENRICHMENT

Notice and Wonder

What do you notice about the sum when you subtract a negative number? Why do you think that is? What does that make you wonder?

TEACHER MOVES

Some students may still need a visual to help with adding/ subtracting integers. Consider having students continue to use either integer chips or number line models to help them.

1.4.4 BRING IT TOGETHER: ADDING AND SUBTRACTING INTEGERS

TEACHER MOVES

For question 1, consider splitting the class into three groups. One group is responsible for statement A, one group is responsible for statement B, and the other is responsible for statement C.

For each statement, have each group complete the statement as a team and have each team share their answer and determine, as a class, if everyone agrees with that statement.

Prompt groups to provide an example for each statement.

This discourse will support their approach in 1.4.6 Try It.



1.4.4 Bring it Together: Adding and Subtracting Integers

1. Complete the statements.

a. When adding integers of the same sign,

☐ add
☐ subtract

the absolute values of the integers.

The sum will have the ☐ same
☐ opposite sign as the integers being added.

b. When adding integers of different signs,

☐ add
☐ subtract

the absolute values of the integers.

The sum will have the same sign as the integer with

the ☐ larger
☐ smaller absolute value.

c. When subtracting integers, create an equivalent

expression by changing the expression to

☐ addition
☐ multiplication

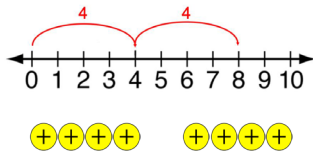
of

☐ a zero pair.
☐ the additive inverse.



1.4.5 Guided Practice: Multiplying and Dividing Integers

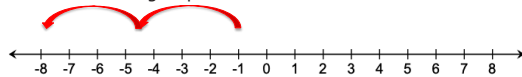
The expression 2×4 means 2 groups with 4 units in each group. This can be represented on the number line or using integer chips as shown.



1. The expression $2 \times (-4)$ is given.
 - a. Create 2 groups of -4 using the integer chip model.



- b. Model 2 groups of 4 units to the left of 0.



- c. Complete the equation.

$$2 \times (-4) = \underline{-8}$$

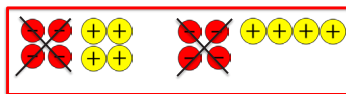
- d. Explain how the expressions $(-2) \times 4$ and $2 \times (-4)$ differ.
The first expression shows 4 groups of -2 while the second expression shows 2 groups of -4 .

2. Complete the visual rule for multiplication of positive and negative numbers.

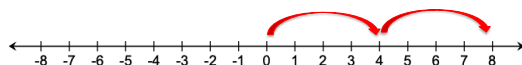
Visual Model of Multiplication	Sign of Product
 $\times$ 	Negative 
 $\times$ 	Negative 

3. Complete the statement.
 The expression $(-2)(-4)$ indicates that 2 groups of -4 should be **removed**.
4. Consider how to represent the expression $(-2) \times (-4)$.

- a. Create an integer chip model to represent this expression.



- b. Model the expression on the number line.



- c. Complete the statement.

$$(-2) \times (-4) = \underline{8}$$

1.4.5 GUIDED PRACTICE: MULTIPLYING AND DIVIDING INTEGERS

DIFFERENTIATE INSTRUCTION

Consider having students use actual integer chips again to model the multiplication.

DIFFERENTIATE INSTRUCTION

Discussion provides additional entry points for auditory learners. Consider utilizing sentence stems and provide ample time for students to grapple with academic language:

The expression $(-2) \times 4$ and $2 \times (-4)$ are different because _____; this makes me wonder _____

ENRICHMENT

Does the order matter when multiplying? What is the property that confirms that order does not matter when multiplying?

TEACHER MOVES

Students may struggle modeling this expression with their integer chips. Notice students who are struggling and consider practicing modeling a few more expressions as a class.

QUESTIONING

Why is multiplying a negative by a negative going to give you a positive answer?

1.4.5 GUIDED PRACTICE: MULTIPLYING AND DIVIDING INTEGERS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider posting a visual of the rules for adding, subtracting, multiplying and dividing integers for students to reference.

ENRICHMENT

What does it mean to have an inverse relationship?

TEACHER MOVES

Consider having students create a visual rule for division of positive and negative numbers.

5. Complete the visual rule for the multiplication of two negative numbers.

Visual Model of Multiplication	Sign of Product
	Positive 

6. Multiplication and division have an inverse relationship. Use this relationship to complete the table.

Given Equation	Can Be Thought of As ...	Related Equation
$3 \times \underline{7} = 21$		$21 \div 3 = \underline{7}$
$-8 \times \underline{-7} = 56$		$56 \div (-8) = \underline{-7}$
$20 \div \underline{-4} = -5$		$(-5)(\underline{-4}) = 20$
$-\left(\frac{36}{4}\right) = \underline{-9}$		$\underline{-9} \times 4 = -36$

7. How are the rules for multiplication of integers related to the rules for division of integers?
The rules for multiplication and division of integers are the same.

**1.4.6 Try It: Integer Operations**

1. Evaluate each expression.

a. $34 + (-21)$

13

b. $-67 + 40$

-27

c. $-333 + (-509)$

-842

d. $700 + (-814)$

-114

e. $632 - (-108)$

740

f. $942 - 621$

321

g. $-78 - (-324)$

246

h. $105 - (-75)$

180

i. $-5 \times (-6)$

30

j. $-22 \div (-11)$

2

k. $-24 \div 3$

-8

l. $9 \times (-4)$

-36**1.4.6 TRY IT: INTEGER OPERATIONS****TEACHER MOVES**

Notice if students are forgetting the negative sign after completing the operation.

For example, students who answer “27” for question 1b demonstrate a correct understanding of subtracting the integers, but either forgot to include the sign or do not understand how to determine if the answer is positive or negative.

Prompt these students to utilize questioning techniques below as a self-check when performing operations.

DIFFERENTIATE INSTRUCTION

Using integer chips and number line models will not be an effective method when the integers being used are so large. Consider substituting out some of the problems for smaller integers that students can use integer chips or number lines to help model.

**1.4.7 Wrap-Up: Reflection**

1. In this lesson, multiple concepts about operations with integers were reviewed. Rate yourself on the following skills based on your level of confidence.

Statement	Rating
	1 – I struggle with this. 2 – I have some skills. 3 – I am almost there. 4 – I have got this!
I can add integers.	<i>Answers will vary.</i>
I can subtract integers.	<i>Answers will vary.</i>
I can multiply integers.	<i>Answers will vary.</i>
I can divide integers.	<i>Answers will vary.</i>

1.4.7 WRAP-UP: REFLECTION**TEACHER MOVES**

Establish with your students that their honest reflection is valued and plays an important role in your teaching. Consider creating a chart for the class to monitor which operations may need further support. Operations with integers are considered prior knowledge for grade 7. Consider how to remediate students whose confidence on operations with integers is rated as a 1 or a 2.

DIFFERENTIATE INSTRUCTION

Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.